AP CALCULUS AB

Unit 1 • Lesson 1.11 notes

Lesson 1.11 • Defining Continuity at a Point

Lesson notes • Learning goals, key ideas, worked examples, common mistakes, quick check

Learning goals

- State the three conditions for continuity at a point.
- Justify continuity or discontinuity with numbers, as required on the AP exam.

Vocabulary. continuous at a point, definition of continuity.

Definition

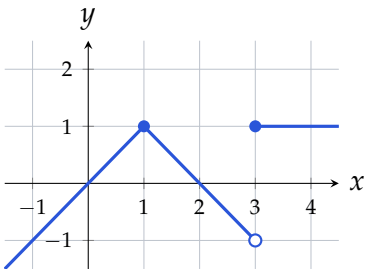
f is continuous at $x = c$ when (1) $f(c)$ is defined, (2) $\lim_{x\to c} f(x)$ exists, and (3) $\lim_{x\to c} f(x) = f(c)$.

Example 1 • A complete justification

$f(x) = \begin{cases} x^2 + 1, & x \le 2 \\ 4x - 3, & x > 2. \end{cases}$ Is f continuous at $x = 2$?

Solution. $f(2) = 5$. $\lim_{x\to 2^-} f(x) = 5$ and $\lim_{x\to 2^+} f(x) = 8 - 3 = 5$, so $\lim_{x\to 2} f(x) = 5$. Since $\lim_{x\to 2} f(x) = f(2)$, f is continuous at $x = 2$.

Example 2 • From a graph



Solution. At $x = 1$: $f(1) = 1$ and both one-sided limits are 1: continuous (a corner does not break continuity). At $x = 3$: left limit -1, right limit 1: condition (2) fails, so f is not continuous at $x = 3$.

Common mistakes

- Checking only that the two pieces "meet" without checking the value $f(c)$.
- Writing "the graph has no holes" as a justification; the AP exam expects the three conditions with values.

AP exam tip

Model sentence: "$\lim_{x\to 2^-} f(x) = 5 = \lim_{x\to 2^+} f(x)$, so $\lim_{x\to 2} f(x) = 5 = f(2)$. Therefore f is continuous at $x = 2$."

Quick check

1. Which condition fails for $f(x) = \frac{x^2-1}{x-1}$ at $x = 1$?
2. $g(3) = 7$ and $\lim_{x\to 3} g(x) = 7$. Is g continuous at 3?

Answers. 1. Condition (1): $f(1)$ is undefined. 2. Yes.